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ORIGINAL PAPER

A potentially probiotic strain of *Enterococcus faecalis* from human milk that is avirulent, antibiotic sensitive, and nonbreaching of the gut barrier

Jasia Anjum[1,2] · Arsalan Zaidi[1,2] · Kim Barrett[3] · Muhammad Tariq[1,2]

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Abstract

Human milk is a key source of promising probiotic lactic acid bacteria. The *Enterococcus* species, because of their dual commensal and pathogenic nature, demand critical safety analysis to establish them as probiotic candidates. In this study, eighteen *E. faecalis* strains from human milk of mothers living in Pakistan were typed at the strain level by riboprinting. The typed strains were then evaluated in vitro for physiological safety and the presence of transmissible antibiotic resistance genes, adhesion genes, biogenic amines, and virulence factors. Selected strains were then checked for tolerance to gastrointestinal acid and bile as criteria for probiotic efficacy. Molecular typing revealed that the strains fell into five distinct clusters or ribotypes. Testing revealed that they were non-hemolytic; however, all strains had gelatinase activity except NPL-493. The isolates were susceptible to most clinically important antibiotics except streptomycin. Molecular screening for antibiotic resistance genes, adhesion genes, biogenic amines, and virulence factors indicated that none of the strains possessed resistance genes for aminoglycosides, vancomycin, bacitracin, tetracycline, or clindamycin. Most virulence factors were absent except for the genes *gelE* and *efaAs* associated with gut adhesion and translocation, which were present in all except NPL-493. Strain NPL-493 was the most promising probiotic candidate demonstrating significant tolerance to the acid, bile, and digestive enzymes in the human GIT and antibacterial activity against multiple pathogens. The study concluded that *E. faecalis* NPL-493 from human milk was safe among all the strains and could be considered a potential probiotic.

Keywords *E. faecalis* · Human milk · Virulence · Translocation · Probiotics

Abbreviations

EFSA	European food safety authority
GIT	Gastrointestinal tract
BSH	Bile salt hydrolase
PCR	Polymerase chain reaction
MIC	Minimum inhibitory concentration
MRS	DeMan, rogosa sharpe

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✉ Arsalan Zaidi
azlanzaidi@yahoo.com

1 National Probiotic Lab, National Institute for Biotechnology and Genetic Engineering-College (NIBGE-C), Jhang Road, Faisalabad 38000, Punjab, Pakistan

2 Pakistan Institute of Engineering and Applied Sciences (PIEAS), Nilore 45650, Islamabad, Pakistan

3 Department of Medicine, University of California San Diego, San Diego, CA 92093-0063, USA

Background

Breast milk constitutes a holistic dietary source for infants that provides complete nutrition, confers protection against various infections, and establishes the foundation of their gut microbiota (Fernández et al. 2013). The enterococci are a part of the core autochthonous microbiota (Reviriego et al. 2005), and several species such as *E. faecalis* and *E. faecium* are also found in the adult GIT. Probing new and underexplored niches for probiotic enterococci is of scientific interest since it provides new clues on the extent of probiotic potential in the enterococci (Bagci et al. 2019). The presence of *E. faecalis* strains in human milk has been reported in many studies, and some have demonstrated probiotic potential (Huang et al. 2019; Khalkhali and Mojgani 2017; Kvist et al. 2008; Reis et al. 2016; Taweerodjanakarn et al. 2019; Toğay et al. 2014). However, the dual nature of enterococci as candidate probiotics and opportunistic pathogens remains controversial (Franz et al. 2011). Enterococci as probiotics can prevent various human and animal diseases, alleviate the symptoms of irritable bowel syndrome and antibiotic-induced diarrhea, and prevent some chronic intestinal diseases (Hanchi et al. 2018). Despite their probiotic promise, concerns about their safety, especially those related to virulence and antibiotic resistance, have hindered the broader application of enterococcal probiotics (Hanchi et al. 2018; Jahan et al. 2015). There has been evidence of virulence factors and related genes in *E. faecalis* from human milk (Khalkhali and Mojgani 2017; Toğay et al. 2014), but virulence studies of enterococci are complex because several genes are involved, and the mechanism of pathogenicity is not well understood (Comerlato et al. 2013).

There has been a lack of studies from Pakistan on the probiotic potential of *E. faecalis* strains originating from human milk and the prevalence of virulence factors in such strains. Studies from other countries reported that *E. faecalis* strains from human milk could harbor virulence genes, but their role in translocation across the intestinal epithelial barrier is unknown (Santana et al. 2020). Similarly, the effectiveness of *E. faecalis* in preventing irritable bowel syndrome and diarrhea and lowering cholesterol has been investigated in very few studies (Christoffersen et al. 2012; Franz et al. 2011; Yamaguchi et al. 2013).

Our work has determined the incidence of virulence factors in *E. faecalis* strains from human milk of mothers indigenous to central Pakistan and evaluated their potential for testing as candidate probiotics.

Methods

Isolation and identification of strains

Bacteria and growth conditions

Eighteen isolates of *E. faecalis* were obtained from the National Probiotic Lab, NIBGE Faisalabad, Pakistan, previously isolated from human milk and identified through 16sRNA sequencing (Table 1) and published (Anjum et al. 2020), whereas the remainder is unpublished data. All the identified sequences were submitted to NCBI Genbank for accession number acquisition (Fig. 1). Stocks were cultured aerobically in MRS (deMan, Rogosa, and Sharpe broth, Merck, 1.10661.0500, Germany), at 37 °C, and log-phase cultures were used for all experiments.

Phylogenetic analysis and strain mapping

Phylogenetic analyses of the isolates were conducted using MEGA7, and the subspecies strain-level identification was made using an automated restriction fragment length polymorphism (RFLP) technique called riboprinting (Clark 1997).

Riboprinting involves an rRNA probing method because bacterial rRNA operons have highly conserved genes flanked by variable DNA regions resulting in different RFLP bands named ribotypes. Ribotyping enables strain analysis without prior genomic DNA sequence knowledge and efficiently generates reproducible results (Li et al. 2009). Following the manufacturer's instructions; Ribotyping was performed on a RiboPrinter microbial characterization system (Hygiena, UK). Purified single colonies were grown to log-phase and analyzed using the RiboPrinter. Strains with > 85% similarity to a known pattern in the RiboPrinter database were considered a match.

Safety assessment in vitro

Gelatinase activity

Overnight log-phase cultures of selected strains in MRS broth were diluted in MRS broth to an OD_{630} of 0.07 and streaked on nutrient agar plates containing 3% (w/v) gelatin. Plates were incubated at 37 °C for 48 h and then held at 4 °C for 4 h. Hydrolysis of gelatin was assessed by measurements of the opaque halos around the colonies (Gómez et al. 2016). A type strain of *Staphylococcus aureus* (ATCC 25,923, Microbiologics Inc.) was used as a positive control.

Hemolytic activity

Overnight log-phase cultures of selected strains in MRS were adjusted to an OD_{630} of 0.07 and streaked on blood agar plates and incubated aerobically at 37 °C for 24 h. A type strain of *Streptococcus pyogenes* (ATTC 25,923) was used as a positive control. Transparent or opaque zones around colonies indicate full hemolysis and greenish-yellow zones of partial hemolysis.

Antimicrobial susceptibility

Log-phase cultures of the test bacterial strains were adjusted to OD_{630} of 0.07, and 100 µl aliquots were spread on MRS plates. E-test strips (BioMérieux, France) for ampicillin, clindamycin, gentamicin, erythromycin, streptomycin, tetracycline, vancomycin, kanamycin, and chloramphenicol were aseptically placed on the surface of the dried plates, which were incubated at 37 °C for 48 h. All the strips used carried antibiotics ranging in concentration from 0.016 to 256 µg/mL except streptomycin, which had 0.0 64 to 1024 µg/mL. Each strain's minimum inhibitory concentration (MIC) was determined by noting where the clear elliptical zone around the strip intersected it (Georgieva et al. 2015).

Table 1 Basic profile of *E. faecalis* strains isolated from human colostrum

Sr. no	Sampling place	Mode of delivery	Mother age (year)	Dairy product	#Disease	#Antibiotic	Isolate code	Gram stain	Catalase	&Molecular Identification	Riboprint ID	Accession no	Reference
1	FSD	NR	28	Milk	No	Metronidazole	NPL-154	+ cocci	–	*E. faecalis*	6216	MW835161	This study
2	FSD	CS	27	Milk, yogurt	No	Metronidazole	NPL-163	+ cocci	–	*E. faecalis*	6216	MW835162	This study
							NPL-165	+ cocci	–	*E. faecalis*	16,162	MW835163	This study
3	FSD	CS	26	Milk, yogurt	No	Metronidazole	NPL-168	+ cocci	–	*E. faecalis*	6216	MW835164	This study
							NPL-169	+ cocci	–	*E. faecalis*	6216	MW835165	This study
4	FSD	CS	35	Milk, yogurt	No	Metronidazole	NPL-178	+ cocci	–	*E. faecalis*	6216	MW835166	This study
							NPL-180	+ cocci	–	*E. faecalis*	6217	MW835167	This study
5	FSD	CS	27	Milk, yogurt	No	Metronidazole	NPL-183	+ cocci	–	*E. faecalis*	6217	MW835168	This study
							NPL-185	+ cocci	–	*E. faecalis*	6217	MW835169	This study
							NPL-186	+ cocci	–	*E. faecalis*	6217	MW835170	This study
6	FSD	NR	32	Milk	No	No	NPL-151	+ cocci	–	*E. faecalis*	6217	MK038954	(Anjum et al. 2020)
							NPL-153	+ cocci	–	*E. faecalis*	6217	MW835160	(Anjum et al. 2020)
7	FSD	NR	27	Yogurt	No	No	NPL-148	+ cocci	–	*E. faecalis*	6217	MK038951	(Anjum et al. 2020)
8	FSD	NR	30	Milk, yogurt	No	No	NPL-160	+ cocci	–	*E. faecalis*	6217	MK038958	(Anjum et al. 2020)
9	LHR	CS	25	Yogurt	No	Metronidazole	NPL-860	+ cocci	–	*E. faecalis*	16,157	MW835172	This study
10	FSD	CS	30	Milk	No	Metronidazole	NPL-146	+ cocci	–	*E. faecalis*	16,162	MW835157	This study
							NPL-147	+ cocci	–	*E. faecalis*	6216	MW835158	This study
11	ISB	CS	27	Milk, yogurt	No	Metronidazole	NPL-493	+ cocci	–	*E. faecalis*	*	MW835171	This study

FSD Faisalabad (District headquarter hospital, Allied hospital, Umer hospital), *LHR* Lahore general Hospital, *ISB* Islamabad (Nescom hospital Islamabad)

CS indicates Cesarean mode of delivery, *NR* indicates normal vaginal delivery

#Disease and Antibiotic intake refers to any infectious and non -infectious disease that the mother has been diagnosed with and the medicines being taken at the time of sampling

$ colostrum indicates milk collected within two to four days after birth

& Molecular identification means 16 s RNA sequencing of isolates

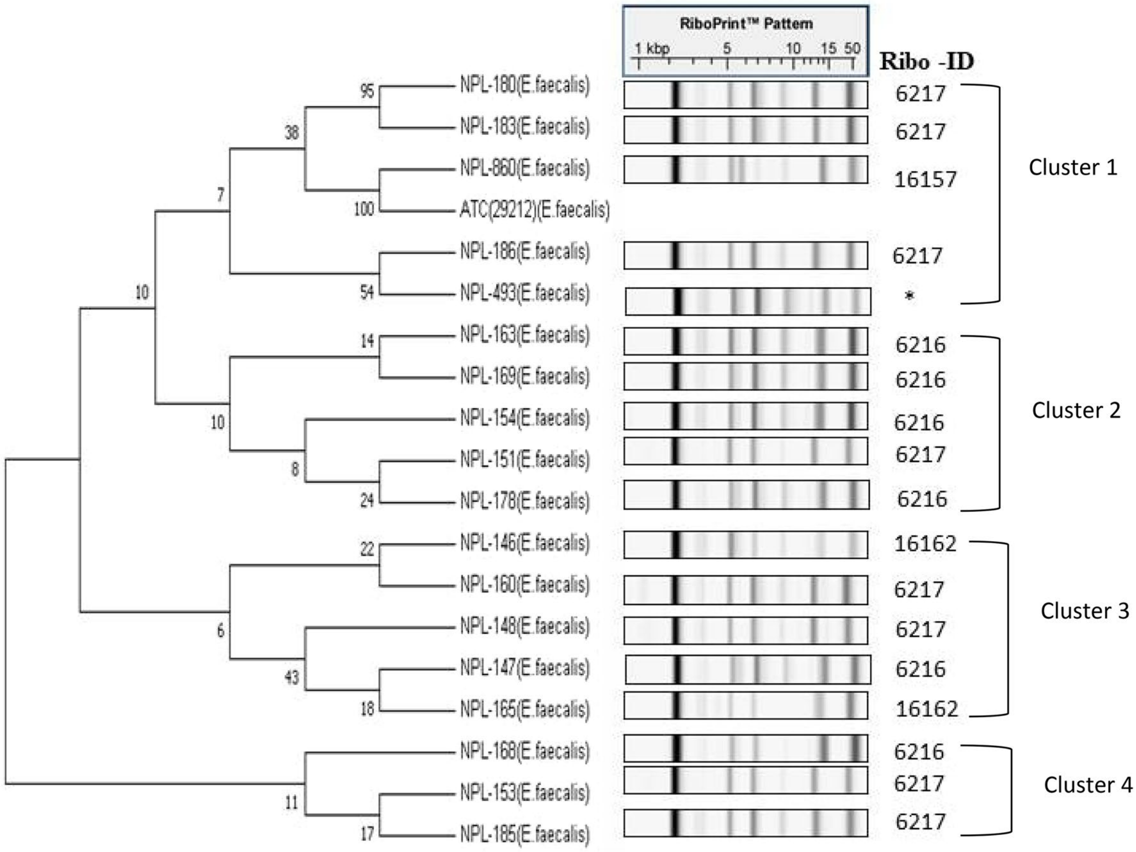


Fig. 1 Evolutionary relationships of *E. faecalis* strains isolated from human milk using Neighbor-Joining method in MEGA7 software. The bootstrap consensus tree inferred from 1000 replicates is taken to represent the evolutionary history of the taxa analyzed

Biogenic amine production

The capacity of the strains to generate biogenic amines was checked using a published method (Belicová et al. 2013). Briefly, a decarboxylation of amino acids was observed on a decarboxylating medium (Sigma-Aldrich, D2935-500G) containing 2% L-tyrosine (Alfa Aesar, Cat. no. J61770), L-histidine (Sigma-Aldrich, Cat. no. 41932), and L-ornithine (Sigma-Aldrich, Cat. no. O6503). The strains were subcultured twice in decarboxylating medium supplemented with one of the three amino acid precursors described above at (1 g/L) and incubated at 37 °C for 48 h. The broth culture was spotted on decarboxylating medium plates containing the respective amino acids and incubated at 37 °C for 48 h. Amine production was scored as a purple halo around the streak line except for media containing tyrosine, where a clear halo around the streak line indicated a positive result. Experiments were performed in triplicate.

Adhesion and translocation studies

Growth and preparation of T84 cell monolayers The human adenocarcinoma epithelial cell line, T84 (passage 19–30), was grown in 1:1 DMEM/F-12 (Gibco™, Cat: 11330-032) supplemented with 5% newborn calf serum (NCS) and antibiotics (penicillin/streptomycin, 1% (v/v) (Gibco™, 15,140,122). The cells were initially passaged in tissue culture flasks, and after reaching 70–80% confluence, were detached with 0.25% trypsin (Gibco™ 25,200–056) and seeded on transwell filters (12-mm diameter; 0.4um pore size, polystyrene membrane; Corning Costar, NY, Cat. no. 3460) at a density of 5×10^5 cells/insert.

Adhesion of *E. faecalis* strains to T84 cell monolayers For adhesion assays, we followed established protocols (Cebrián et al. 2012; Sartingen et al. 2000). T84 cells were cultured in DMEM/F-12 (1:1) (Dulbecco's modified Eagle's medium with Ham's F-12, Gibco, Cat. no. 11330-032) supplemented with 5% NCS and antibiotics (penicillin/streptomycin, 1% v/v), seeded into culture dishes, and grown to a confluent monolayer. One day before the experiment, the DMEM-F12 medium was replaced with an antibiotic-free medium. The next day, a suspension of 1×10^8 CFU of a candidate bacterial strain per ml in infection medium (DMEM-F12 without antibiotics) was added to each tissue culture well, yielding a multiplicity of infection of 100:1. After incubation for 1 h at 37 °C in 5% CO_2, the monolayers were washed four times with PBS to remove nonadherent bacteria and lysed by adding 1 ml of 1% Triton X-100 (9036-19-5, Merck, Germany) for 5 min. The total number of adherent bacteria was determined by plating aliquots of cell lysates on MRS agar. The data are expressed as means and standard error of means for three independent assays.

Translocation of *E. faecalis* strains across T84 cell monolayers Translocation of bacteria was evaluated using the T84 human adenocarcinoma epithelial cell line (Zeng et al. 2004). From passages 19 to 30, 1×10^5 T84 cells were seeded on transwell filters (12-mm diameter; 0.4 µm pore size, polystyrene membrane; Corning Costar, NY, Cat. no. 3460). The establishment of the monolayer was monitored by measuring the transepithelial resistance (TEER) using an STX2 chopstick electrode set (EVOM^2, World Precision Instruments, USA). Once the cells were confluent on the membrane (TEER > 1500 $\Omega \cdot cm^{-2}$), the medium was replaced with antibiotic-free DMEM/F-12 one day before the experiment.

For the translocation experiment, log-phase bacterial cultures were inoculated into fresh MRS broth and incubated at 37 °C for about 3 h to an OD_{630} of 0.4. The bacterial cells were collected by centrifugation at 3500 g, and after washing the pellet with PBS, the titer was adjusted to 5×10^8 CFU/ml in DMEM/F-12 supplemented with 10% NCS without antibiotics. On the experiment day, the filters were washed twice with DMEM/F-12 medium supplemented with 10% NCS without antibiotics. Fresh medium (1 ml plus 10% NCS with no antibiotics) was added to the lower chamber, and about 8×10^7 bacteria in 400 µl of DMEM/F-12 was added to the upper chamber. TEER was monitored for up to 5 h. Viable bacteria in the lower chambers were counted at 0, 4, and 6 h by removing 100-µl aliquots, serially diluting them, and plating on MRS agar plates; viable bacteria in upper chambers were similarly counted at 6 h. For each strain, three transwells were used, and the experiments were repeated at least three times. To validate the T84 epithelial barrier model, we also used *Salmonella enterica* subsp. *enterica* serovar Typhimurium (ATCC SL1344) as a positive control and untreated T84 cells as a negative control.

Molecular analysis of *E. faecalis* strains for virulence genes and genes encoding antibiotic resistance and biogenic amines

The *E. faecalis* strains were evaluated for the presence of genes for virulence, antibiotic resistance, and amino acid decarboxylation (Casarotti et al. 2017). DNA was extracted using the Gene Jet genomic DNA kit (K0721, Thermo-Scientific, USA), and DNA concentration was measured by NanoDrop 2000C (Thermo Scientific) and gel electrophoresis. PCR (T100, BioRad, USA) was performed using the Phusion high-fidelity PCR master mix (Thermo-Scientific USA, Cat. no. F-532S) and published primer sets, and amplification protocols as specified (Table S1). The type strain *E. faecalis* ATCC 29,212, was used as a control for PCR. The amplicons were analyzed by 1.2% (w/v) agarose gel electrophoresis.

Evaluation of probiotic properties

Growth dynamics and acid production in aerobic vs. anaerobic environments The following protocol determined growth rate and acid production. The initial OD_{630} was adjusted to 0.07 for aerobic cultures and 0.1 MacFarland units (MFU) for anaerobic cultures (Anjum et al. 2020) in 5 ml of MRS broth (Merck, Germany). The cultures were incubated aerobically for 24 h at 37 °C in an incubator (IFA-110–8, ESCO, Singapore) and anaerobically in an atmosphere of 3% H_2, 5% CO_2, and 92% N_2 (Bactron 300, Shell Lab, USA). Growth was monitored every 2 h, and the pH was measured simultaneously to monitor acid production.

Gastric acid tolerance test For acid tolerance, a published method was used (Bouridane et al. 2016) with modifications. Log-phase cultures were harvested by centrifugation (2268 g, 4 °C, 5 min), pellets were washed twice and resuspended in sterile PBS. The OD_{630} of cultures was adjusted to 0.07 with PBS at a pH of 2.8, 4.2, or 5.7 and incubated aerobically at 37 °C for 3 h. Cultures of test strains at an initial OD_{630} of 0.07 buffered to pH 5.7 were used as controls. Following 3 h of acid exposure, 100 µl of stressed and control culture were transferred to 10 ml of fresh MRS_c broth and growth checked for 5 h by measuring OD_{630} compared to that at pH 5.7.

Bile tolerance test Log-phase cultures were harvested by centrifugation (2268 g, 4 °C, 5 min), and pellets were washed and resuspended in PBS. The cultures were added to fresh MRS broth containing 0.0%, 0.1%, 0.2%, and 0.3% (w/v) ox gall (Oxoid, UK) at a starting OD_{630} of 0.07. The cultures were incubated aerobically at 37 °C for 6 h (Bouridane et al. 2016), and endpoint OD_{630} was determined. Results were compared to cultures grown in MRS without bile.

Bile salt hydrolase (BSH) activity BSH activity was assayed using the protocol described by (Sedláčková et al. 2015). Soft agar plates made with MRS broth (52.5 g/l; Oxoid), bacteriological agar (7.5 g/l; Oxoid), porcine bile (0.3% (w/v), B8631, Sigma), and $CaCl_2$ (0.37 5 g/l, Daejung, Korea) were prepared. The plates were incubated at 37 °C for 48 h. Aliquots of different strains (10 µl, 10^6 CFU/mL), cultured aerobically for 12 h, were inoculated into wells made by puncturing the agar surface. Plates were incubated at 37 °C for 72 h under aerobic conditions, and visible halos around the punctures were indicative of BSH activity.

Tolerance to digestive enzymes Selected bacterial strains were grown from stock cultures in freshly prepared MRS, harvested in log phase and used to re-inoculate 10 ml of fresh PBS containing trypsin (m/v) (650,200, Calbiochem) at 0, 0.4, 0.8, 1.2, 1.6 and 2% at a starting OD_{630} of 0.4 and pH 6.8. For assaying pepsin (21,560,021, Bio World, USA) tolerance, 10 ml PBS containing (m/v) at 0, 0.4, 0.8, 1.2, 1.6 and 2% pepsin was inoculated to OD_{630} at 0.4 and pH 3. The cultures were incubated at 37 °C for 3 h. Cultures suspended in PBS to an OD_{630} of 0.4 and having pH 7.5 served as positive controls. After enzyme exposure, the cultures were harvested, resuspended in 3 ml of sterile MRS media, and incubated at 37 °C for another five hours. Growth was measured as ΔOD_{630}, compared to the controls (Lei et al. 2015; Shi et al. 2020).

Antibacterial assays Cell-free supernatants (CFS) were obtained by centrifugation (2268 g, 10 min) of bacterial cultures. The pH of CFS was determined, and equal portions were neutralized to pH 7.0, digested with proteinase K (1 mg/ml) for 1 h at 37 °C, heated 10 min at 60 °C, and 1 h at 100 °C. After heating, CFS was filter-sterilized, and a 1% v/v dilution in PBS was added to wells punctured into soft nutrient agar (M002-500 g, Himedia, India) plates covered with a lawn of 10^6 CFU/mL of one of the indicator strains: *Staphylococcus aureus* (ATCC 19,615), *Streptococcus pyogenes* (ATCC 25,923), or *E. coli* (ATCC 25,922). The plates were kept for 2 h at 4 °C for the CFS to diffuse into the agar and then incubated at 37 °C for 72 h. The diameter of the clear area around each puncture in mm was measured by a Vernier caliper and taken as indicative of antibacterial activity.

Statistical analysis

All experiments were performed in triplicate from independent seedings, with three replicates each. Statistical differences between all groups of treatments for TEER were determined using two-way ANOVA and one-way ANOVA. Differences were considered statistically significant at $P \leq 0.05$, and data are presented as mean + SEM. All statistical analyses were performed using Excel and GraphPad Prism statistical software.

Results

Phylogenetic analysis and strain relatedness

The molecular evolutionary relationship of the strains was phylogenetically mapped, and the variation between the strains was resolved into four different clusters. Riboprinting indicated that the seventeen strains of *E. faecalis* could be clustered in four distinct ribotypes depending on similarity to ribopatterns in the library (< 85%). These ribotypes were assigned DUP IDs DUP-6217, DUP-16157, DUP-16162, and DUP- 6216 corresponding to the different strains of *E. faecalis,* while strain NPL-493 was found to be native to the Pakistani population and stood apart from any reference strain present in the database (Table 1, Fig. 1). These strain-level differences identified by RiboPrinter also confirmed the cluster analysis results indicating the variation in human milk strains representing the non-uniform distribution of different *E. faecalis* strains in the Pakistani population.

Safety analysis of *E. faecalis* strains in vitro

Gelatinase activity

Gelatinase is an extracellular enzyme encoded by the gene *gelE* that digests gelatin (Zielińska and Kołożyn-Krajewska 2018). The presence of large halos around the streak line on agar containing gelatin indicated positive gelatinase activity in all strains except NPL-493 (Figure S1A, Table 3).

Hemolytic activity

The strains did not form any clear zones on blood agar plates, indicating their inability to lyse red blood cells (Table 3, Figure S1C).

Antimicrobial susceptibility assay

Antibiotic resistance is an essential beneficial trait for a candidate probiotic, provided the chromosomally encoded DNA is nontransferable (Gueimonde et al. 2013). In this study, all isolates were susceptible to ampicillin, kanamycin, vancomycin, erythromycin, and tetracycline, while variable resistance and susceptibility were observed for the other antibiotics (Table 2).

Biogenic amine production

All eighteen strains produced tyramine as a biogenic amine, a species-level trait of *E. faecalis*. No activity was observed against ornithine or histidine (Fig S1B, Table 3).

Molecular analysis of antibiotic resistance, virulence factors, and biogenic amines

Investigation of virulence factors by PCR revealed the presence of genes for gelatinase (*gelE*), cell wall adhesin (*efaAs*), and sex pheromone (*cob*) while other supposed virulence genes (*Esp, agg, cpd, ccf, cad, ace, cylM, cylB,* and *cylA*) were not detected. At least four virulence genes were detected in all the *E. faecalis* strains except NPL-493.

Table 2 Phenotypic and molecular antibiotic susceptibility profile of enterococcal strains

Test category	Antibiotic Resistance																
	Macrolides (Erythromycin)			chloramphenicol		Aminoglycosides								β-Lactam Ampicillin		Lincomycin Clindamycin	
	Pheno-type	*ereA*	*ereB*	Pheno-type	*Cat-pIP501*	Phenotype			*aph(2")-Ib*	*aph(2")-Ic*	*aph(2")-Id*	*aph(3')-III-a*	*ant(4')-Ia*	Pheno-type	*β-lactam antibiotics (bla)*	Pheno-type	*lsa*
Strain code						*kanamycin*	*streptomycin*	*gentamycin*									
DUP-16162							✓	✓									
DUP-6216							✓									✓	
DUP-6217							✓										
*DUP-							✓										
DUP-16157							✓									✓	
ATCC-29212								✓									

Test category	Antibiotic Resistance																
	Bacitracin			Tetracyclines Tetracycline							Glycopeptides (Vancomycin)						
Strain code	*bcrB*	*bcrD*	*bcrR*	Pheno-type	*tet(K)*	*tet(O)*	*tet(S)*	*tet(L)*	*tet(M)*	*int-Tn*	Pheno-type	*VAN-A*	*VAN-B*	*VANC1*	*vanC-1*	*vanC-2*	*vanC-3*
DUP-16162																	
DUP-6216																	
DUP-6217																	
*DUP-																	
DUP-16157																	
ATCC-29212				✓													

DUP-16162 (NPL-146, NPL-165); DUP-6216 (NPL-147, NPL-154, NPL-163, NPL-168, NPL-169, NPL-178); DUP-6217 (NPL-148, NPL-151, NPL-153, NPL-180, NPL-183, NPL-185, NPL-186); DUP-* (NPL-493); DUP-16157(NPL-860)

"✓" indicates positive results

Table 3 virulence profile of enterococcal strains

Test category	Virulence genes																		
	Gelatinase					Aggregation substance	Enterococcal surface protein	Hemolysis		Translocation		Sex pheromones				Adhesion of collagen	Serine protease	Hyaluronidase
Strain code	Phenotype	Quorum sensing gene (fsrA)	Quorum sensing gene (fsrB)	Quorum sensing gene (fsrC)	Gelatinase (gelE)	Aggregation substance (asa1)	Enterococcal surface protein (esp)	Phenotype	Cytolysin (cylA)	Phenotype	Endocarditis Antigen (efaAs)	Phenotype	Conjugation gene (cob)	Conjugation gene (ccf)	Conjugation gene (cpd)	Adhesion of collagen (ace)	Serine protease (sprE)	Hyaluronidase (hyl)
DUP-16162	✓		✓		✓					✓	✓		✓					
DUP-6216	✓		✓		✓					✓	✓		✓					
DUP-6217	✓		✓		✓					✓	✓		✓					
*DUP				✓									✓					
DUP-16157	✓		✓		✓					✓	✓		✓					
ATCC-29212	✓	✓	✓	✓	✓					✓	✓		✓					

Test category	Biogenic amines							Adhesion, aggregation, and colonization genes							
	Tyrosine		Histidine			Ornithine									
Strain code	Phenotype	Tyrosine decarboxylase (tdc)	Phenotypic	Histidine decarboxylase (hdc1)	Histidine decarboxylase (hdc2)	Phenotype	Ornithine decarboxylase (odc)	Phenotype	Adhesion proteins (Mub)	Adhesion proteins (mapA)	Elongation Factor (EF-Tu)	Choline binding protein (EF2662-cbp)	Fibrinogen binding protein (EF1249-fbp)	Zinc metalloprotease (EF2380-maz)	Surface protein (prgB)
DUP-16162	✓	✓						✓						✓	
DUP-6216	✓	✓						✓						✓	
DUP-6217	✓	✓						✓						✓	
*DUP	✓	✓						✓						✓	
DUP-16157	✓	✓						✓						✓	

Test category	Biogenic amines							Adhesion, aggregation, and colonization genes							
	Tyrosine		Histidine			Ornithine									
Strain code	Pheno-type	Tyrosine decar-boxylase (*tdc*)	Pheno-typic	Histidine decar-boxylase (*hdc1*)	Histidine decar-boxylase (*hdc2*)	Pheno-type	Ornithine decar-boxylase (*odc*)	Pheno-type	Adhesion proteins (*Mub*)	Adhesion proteins (*mapA*)	Elonga-tion Factor (*EF-Tu*)	Choline binding protein (*EF2662-cbp*)	Fibrino-gen binding protein (*EF1249-fbp*)	Zinc metal-loprotease (*EF2380-maz*)	Surface protein (*prgB*)
ATCC-29212	√	√						√						√	

DUP-16162 (NPL-146, NPL-165); DUP-6216 (NPL-147, NPL-154, NPL-163, NPL-168, NPL-169, NPL-178); DUP-6217 (NPL-148, NPL-151, NPL-153, NPL-180, NPL-183, NPL-185, NPL-186)

*DUP (NPL-493); DUP-16157(NPL-860)

"√" indicates positive results

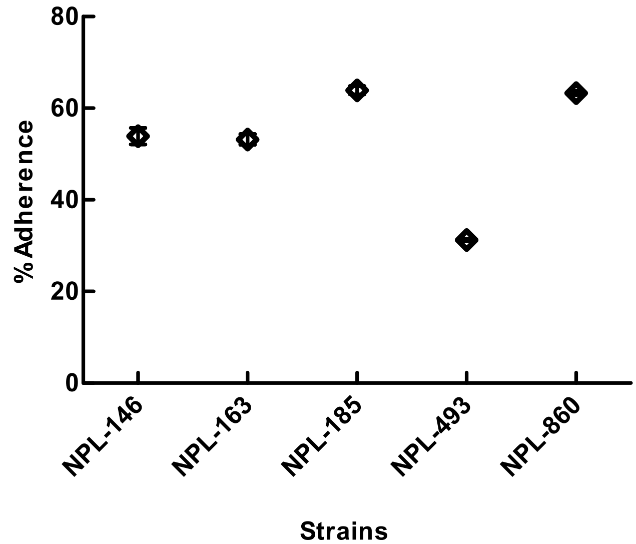


Fig. 2 Adhesion of *E. faecalis* strains to T84-cell monolayers: data are mean ± SEM

Vancomycin resistance (Van^R) is a significant concern and different phenotypes of Van^R exist in enterococci (*vanA, vanB, vanC1, vanC2,* and *vanC3*). The *vanA* phenotype has high-level inducible resistance to vancomycin and teicoplanin. The *vanB* phenotype (encoded by two *vanB* genes) has moderate to high resistance to vancomycin only. The *vanC* phenotype, encoded by two *vanC* genes, demonstrates a non-inducible low-level Van^R. The magnitude of Van^R depends on the expression of different phenotypes. None of these strains was positive for all vancomycin phenotypes. *E. faecalis* strains indicated the absence of lysine and ornithine and the presence of tyrosine decarboxylase genes**.** The PCR results are shown in Tables 2 and 3.

Adhesion and translocation studies of *E. faecalis* strains

Adhesion of strains to T84 cells

Adhesion is pivotal for any would-be probiotic bacterium because it must adhere to intestinal cells for colonization inside the gut. Figure 2 shows that four isolates (NPL-146, NPL-163, NPL-185, NPL-860) have remarkably high adherence (> 50%) to T84 cells, while NPL-493 had relatively weak adherence (< 50%).

Translocation of *E. faecalis* across T84 monolayers

Five representative strains were used in the translocation model (four individual experiments; six wells per experiment). Four strains had similar numbers in the upper chambers after 6 h (about 10^8 at the end of the experiments), and all except NPL-493 were detected in the lower chambers. Except for NPL-493, the strains NPL-146, NPL-185, NPL-163, and NPL-860, were able to pass through the 0.4 µm filter membrane (Fig. 3A). TEER remained high at 6 h, indicating that T84 monolayers were intact throughout the experiment (Fig. 3B).

Probiotic properties of *E. faecalis* strain NPL-493

Growth behavior

NPL-493 showed the highest growth rate at 2 to 8 h of incubation in aerobic and anaerobic environments with faster growth in the former (Fig. 4). The acid production curves indicate that medium pH decreased from 5.5 to 4.8 in its presence (Fig. 4).

Tolerance to acid and bile, and bile salt hydrolase (BSH) activity

NPL-493 showed excellent survival under simulated GIT acid and bile conditions, being able to survive at pH as low as 2.8 and bile concentrations as high as 0.3% (w/v). The strain also precipitated bile salts, and a clear halo of < 10 mm was observed around the wells indicating good BSH activity (Table 4, Figure S1D).

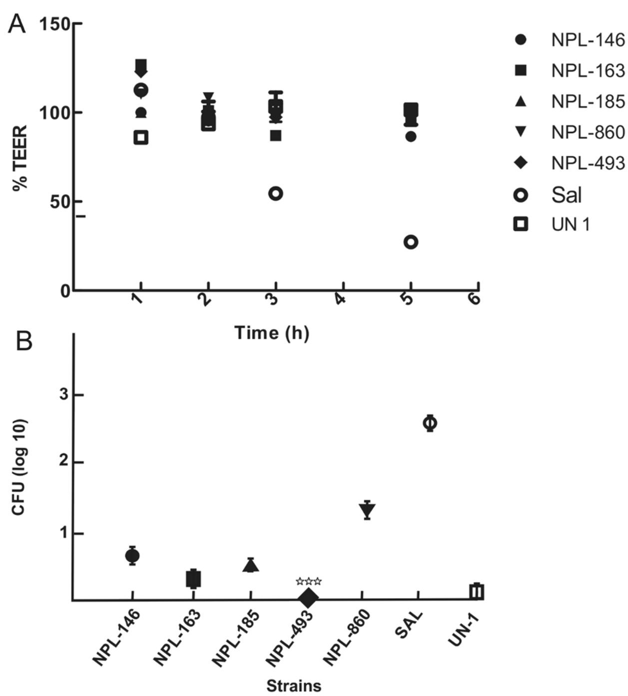


Fig. 3 **a** Comparison of the effect of different strains of *E. faecalis* (NPL-146, NPL-163, NPL-185, NPL-493, NPL-860) on the TEER of T84 human epithelial cell monolayers **b** Translocation of various strains of *E. faecalis* across T84 cells monolayer: SAL is the *Salmonella enterica* serovar Typhimurium, and UN-1 is the untreated T84 cell monolayer. Error bars show the standard deviations of means of independent experiments done in triplicate. Data are expressed as mean ± sd. ***$P \leq 0.0001$ versus control

Tolerance to digestive enzymes

It is crucial to show resistance to digestive enzymes for the survival of a candidate probiotic strain, and growth of NPL-493 was not affected by up to 2% (w/v) of trypsin and pepsin (Table 4).

Antimicrobial activity

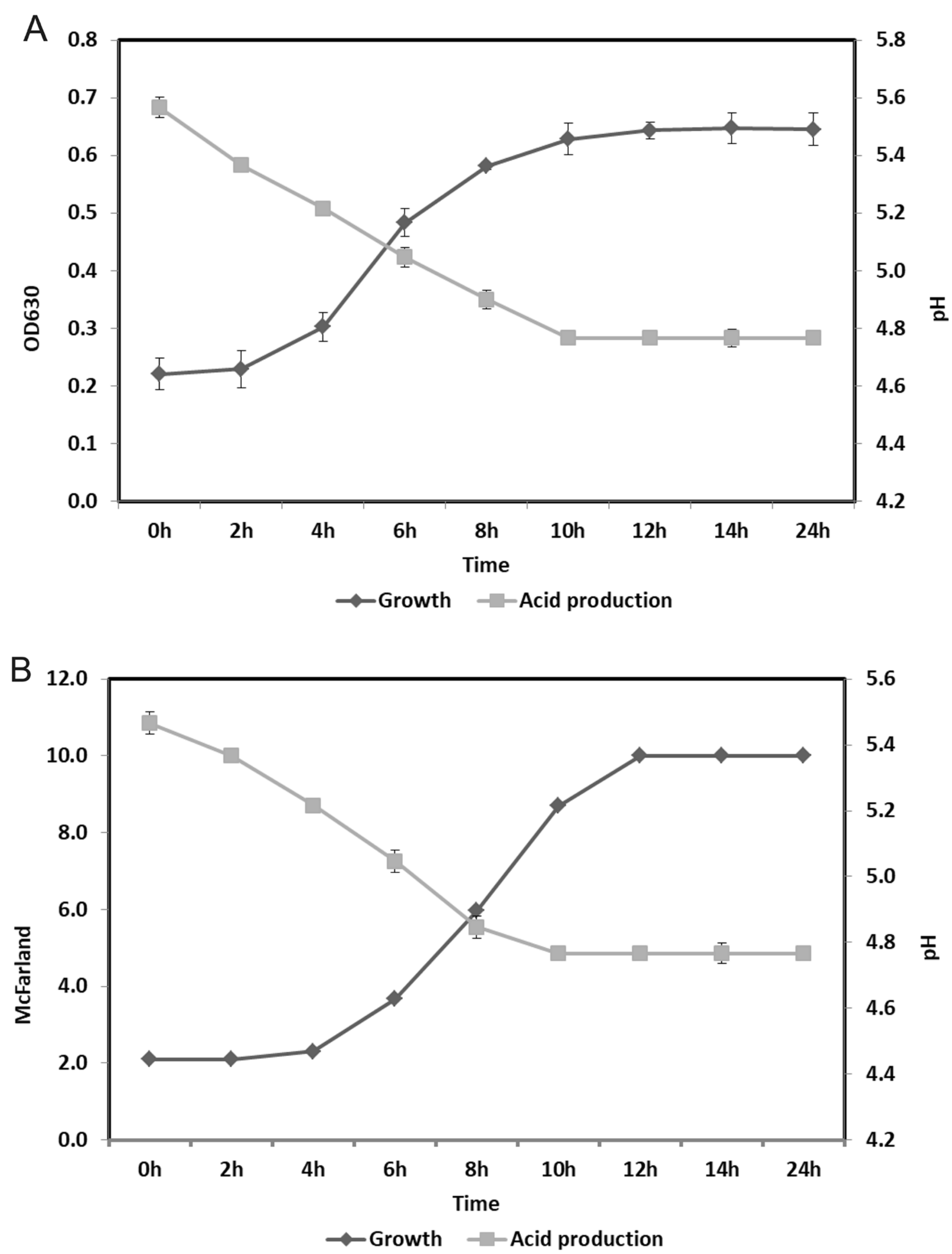
NPL-493 was evaluated for antibacterial activity against *Streptococcus pyogenes*, *Staphylococcus aureus*, *E. coli,* and *S.* Typhimurium. The strain strongly inhibited *S. pyogenes* and showed moderate activity against *E. coli* and *S.* Typhimurium, while no activity was found against *S. aureus* (Table 4, Figure S1E). Proteinase and heat treatment did not influence the antimicrobial activity; however, pH-neutralization of the cell-free supernatant (CFS) eliminated the NPL-493 antibacterial activity, indicating that it was due to acid production.

Discussion

This study provides data on the probiotic and pathogenic potential of *E. faecalis* strains isolated from human milk; focus on virulence potential has been lacking in prior studies on human milk enterococci (Huang et al. 2019; Jiménez et al. 2008; Kvist et al. 2008; Reviriego et al. 2005; Toğay et al. 2014). This aspect should be considered as *E. faecalis,* and *E. faecium* are promising probiotic candidates because of anti-pathogen activity and health-promoting benefits for both adults (Blessing et al. 2020; Hanchi et al. 2018) and neonates (Jiménez et al. 2013). The enterococcal probiotics have not been as studied as *Lactobacillus* and *Bifidobacterium* species (Carasi et al. 2017).

Both *E. faecalis* and *E. faecium* are recurrent species in human milk (Khalkhali and Mojgani 2017), but some studies reported only *E. faecalis* (Huang et al. 2019; Jiménez et al. 2008; Kvist et al. 2008; Toğay et al. 2014) or only *E. faecium* in human milk (Khalkhali and Mojgani 2017; Kıvanç et al. 2016; Reviriego et al. 2005). All the *E. faecalis* strains in this study had a colostral origin (Toğay et al. 2014). Components of the human milk glycome in some mothers allow enterococcal spp. to thrive (Korpela et al. 2018) and low numbers of enterococci correlate with poor neonatal health (Laursen et al. 2020). Their primacy in human colostrum has been demonstrated (Huang et al. 2019; Jiménez et al. 2008). The microbiota of human milk may originate from

Fig. 4 Growth curves and acid production of *E. faecalis* NPL-493 in (**A**) aerobic and (**B**) anaerobic environments



the maternal intestine (Rodríguez 2014), where both of these species abound (Fisher and Phillips 2009), and where *E. faecalis* outnumbers *E. faecium* (Ruiz-Garbajosa et al. 2009) as found here.

Unlike the typical probiotic lactobacilli and bifidobacteria, the enterococci are lacking in global safety certifications such as the qualified presumption of safety (QPS) status from the European Food Safety Authority (EFSA) or Generally Recognized As Safe (GRAS) status conferred by the US Food and Drug Administration (FDA) (Hanchi et al. 2018). Global food safety bodies require defining enterococcal strains as safe before using them as probiotics (EFSA-Panel 2012). For a strain to be approved as a probiotic, it must be unequivocally identified by molecular typing methods (Monteagudo-Mera et al. 2011). *E. faecalis* is a diverse species that has been genotyped to identify strain-level differences (Buhnik-Rosenblau et al. 2013). In this study, the inaccuracy of 16 s rRNA sequencing in differentiating closely related strains was overcome by automated ribotyping, a reliable and reproducible method for strain-level resolution proven to discriminate closely related strains of *E. faecalis* (Arciola et al. 2007; Peta et al. 2006).

The safety risk of using enterococci as probiotics has remained a paramount concern because of virulence factors, antibiotic resistance, and their role in many enterococcal

Table 4 In vitro stress tolerance and probiotic functionality of NPL-493 (*E. faecalis*)

Sr. no	Description of assays	NPL-493
1	Bile resistance (%)	
	0.1	✓
	0.2	✓
	0.3	✓
2	pH resistance	
	2.8	✓
	4.2	✓
	5.7	✓
3	BSH activity	✓
4	Resistance to Trypsin % (m/v)	
	0.0	✓
	0.4	✓
	0.8	✓
	1.2	✓
	2.0	✓
5	Resistance to pepsin % (m/v)	
	0.0	✓
	0.4	✓
	0.8	✓
	1.2	✓
	2.0	✓
6	Antimicrobial activity **	
	S. aureus	−
	E. coli	+
	S. pyogenes	+ +
	S. Typhimurium	+

'✓' represent growth up to 85% of control

**Inhibitory activity represented as follows

− no inhibition

+ zone diameter > 10 mm (Moderate)

+ + > 14 mm (strong)

infections (Arias and Murray 2012; Higuita and Huycke 2014). Thus, profiling potential candidates for antibiotic susceptibility and virulence at the strain level is essential because of the considerable variation (Creti et al. 2004). No study in Pakistan has assessed the prevalence of probiotic and virulence potential of *E. faecalis* strains from human milk, but reports from other countries suggesting that the lower diversity among clinical enterococcal species, including *E. faecalis*, relative to wild strains is ascribable to the virulence factors carried by clinical isolates (Fisher and Phillips 2009). Local studies on *E. faecalis* strains of edaphic and poultry origin revealed a predominance of virulence factors such as *efaAfs*, *gelE*, *sprE,* and *esp* virulence genes and multidrug resistance (Ali et al. 2017; Hasan et al. 2018). However, *E. faecalis* of dairy provenance are generally sensitive to common antibiotics (except methicillin and kanamycin) and are devoid of critical virulence determinants, cytolysin (*cyl*), gelatinase (*gel*), enterococcal surface protein (*esp*), and vancomycin resistance (*vanA* and *vanB*) (Hasan et al. 2018). The human milk *E. faecalis* strains studied here appeared to be like those originating from the soil with a prevalence of *efaAfs* and *gelE* genes. The genes for sex pheromone, *cob*, and cell wall adhesion, *EF2380-maz*, and *efaAfs* were detected in almost all human milk strains, in agreement with (Toğay et al. 2014). It can be argued that the presence of sex pheromone genes may not be directly affecting virulence, yet their expression may favor disseminating antibiotic resistance and virulence determinants carried by pheromone-responsive conjugative plasmids (Inoğlu and Tuncer 2013; Valenzuela et al. 2008). The *EF2380-maz* gene is used in defense and nutrient utilization, which enhances the probiotic potential of LAB strains (de Souza et al. 2019).

The gene *efaAfs* is considered a potential virulence factor as it promotes bacterial adhesion to the human intestinal mucosa. The presence of *efaAs* and *gelE* in enterococci increases their capacity for causing endocarditis in humans (Golob et al. 2019). In this study, all the strains were phenotypically positive for gelatinase function; which also requires several factors besides *gelE* (de Castilho et al. 2019; Tsikrikonis et al. 2012). The product of the *fsr* transcriptional regulator gene is the primary activator of *gelE* expression, and both are important for enterococcal function (Dundar et al. 2015). A mutation or absence of any of the constituents of this operon can result in a gelatinase negative phenotype (Galloway-Pena et al. 2011; Teixeira et al. 2012). In this study, all strains except NPL-493 were positive for *gelE* and *fsrB* and could translocate across T84 cell monolayers, in agreement with previous studies (Zeng et al. 2004). Consistent with past studies, beta-hemolytic activity was not found in any of our strains (Bagci et al. 2019; Valenzuela et al. 2009). Cytolysins are clinically relevant virulence elements (Kayser 2003), and must be absent from any prospective probiotic (Nueno-Palop and Narbad 2011). Their association with pathogenicity has come into question, however, through the discovery of commensal strains of *E. faecalis,* which were more β-hemolytic than those deemed clinical pathogens (Buhnik-Rosenblau et al. 2013), along with some virulence traits such as those related to adhesion that improves probiotic functionality (Nueno-Palop and Narbad 2011).

Safety of enterococci for human use is an overriding concern since many *E. faecalis* strains of human milk origin have been adjudged unsafe, which otherwise exhibited promising probiotic features (Khalkhali and Mojgani 2017). Antibiotic resistance is a natural trait (D'Costa et al. 2011), but its widespread occurrence allows aggressive colonization and incurable superinfections. *Enterococcus* spp. from the milk of healthy mothers possesses resistance to various clinically relevant antibiotics (Huang et al. 2019). The strains here were isolated from mothers who underwent recent C-sections and were on metronidazole which kills Gram-negative bacteria, but enterococci are relatively resistant. These strains were susceptible to most classes of antibiotics, including ampicillin, chloramphenicol, kanamycin, tetracycline, and vancomycin, in line with earlier studies (Jiménez et al. 2008; Toğay et al. 2014). Vancomycin resistance is a worrisome trend in Pakistan (Ejaz 2019), so sensitivity to vancomycin is a desirable bacterial trait. Overuse of macrolides and tetracyclines has resulted in acquiring resistance in enterococci primarily through the horizontal transfer of *ermB* and *tetM* genes (De Leener et al. 2004). *E. faecalis* is intrinsically resistant to clindamycin (a lincosamide) and streptomycin (aminoglycoside) (Hollenbeck and Rice 2012) but was variably present in our strains. Mutations in the *lsaA* gene can result in an inactive efflux pump that removes intracellular clindamycin, rendering the otherwise intrinsically resistant *E. faecalis* susceptible (Sirichoat et al. 2020). The antibiotic susceptibility found here fits more the profile of a classically clinical strain of *E. faecalis* than of the troublesome resistant variants with susceptibility to ampicillin and vancomycin and moderate resistance to aminoglycosides (Kristich 2014). The risk of virulence and AMR genes being transmitted by *Enterococcus* spp. is low (Faron et al. 2016; Jahan et al. 2015). Our *E. faecalis* strain NPL-493 meets EFSA's safety criteria based on susceptibility to ampicillin and absence of virulence markers like *hyl* and *IS16* (EFSA-Panel 2012).

Biogenic amines (BAs) are organic nitrogen compounds produced by lactic acid bacteria by decarboxylation of amino acids. High levels of BAs can cause headache, vomiting, heart palpitations, and diarrhea in humans (Barbieri et al. 2019); however, the low levels of spermidine, putrescine, and spermine in human milk are deemed essential for the normal development of infant gut and immune systems (Perez et al. 2016). Although aminobiogenic capability is reported to be strain-dependent, tyramine and putrescine biosynthesis is a species-level trait in *E. faecalis* (Landete et al. 2018), as also seen here. Synthesis of tyramine is part of the *E. faecalis* acid-response machinery (Perez et al. 2015), but tyramine is a safety concern as it is inimical to human health (EFSA-Panel 2011). Low tyrosine levels still allow enterococcal spp. as food additives (Andrighetto et al. 2001) and probiotics (Li et al. 2018). It is possible to limit aminobiogenesis in the industrial fermentation process (Gardini et al. 2008; Naila et al. 2010), and the commercial track record of *E. faecalis* is remarkably benign (Dubin and Pamer 2018).

After passing the stresses of the upper GIT, the probiotic must adhere to the cells of the colonic mucosa before its beneficial properties can be manifest (Nueno-Palop and Narbad 2011). Since enterococci reside in the colon (James et al. 2020), the T84 cell monolayers act as a good model for mucosal adhesion, giving a better reflection of their adhesion activity in vivo than Caco-2 cells, which are enterocytes from the small intestine (Devriese et al. 2017). The strains isolated here were > 50% adherent to T84 cell monolayers, significantly more than the probiotic strains of *E. faecalis* reported elsewhere (Cebrián et al. 2012) and other *Enterococcus* and LAB spp. (Carasi et al. 2017). The ability to compromise gut barrier integrity is a serious safety risk for a presumptive enterococcal probiotic (Liong 2008). There is evidence that *E. faecalis* can cross the GI barrier (Wells et al. 1990) and cause diseases such as colorectal cancer (Khan et al. 2018). Such a translocation appears to be via an intracellular route rather than through compromised tight junctions between epithelial cells (Archambaud et al. 2019) irrespective of whether or not the translocating cells have a functional gelatinase which has earlier been linked to the degradation of the tight junction protein, E-cadherin, and barrier failure (Steck et al. 2011).

E. faecalis are metabolically more active under aerobic conditions (Portela et al. 2014), which could explain the enhanced survival of NPL-493 under aerobic conditions. A potential probiotic strain must survive stomach acid (pH 2–3), the high bile concentration of the small intestine (0.3%), and the digestive enzymes; enterococci fulfill these criteria well (Hanchi et al. 2018). These GIT stressors act out over a dynamic range; the pH of the gastric juice varies from 2.0 to 3.0, while the bile concentration in the small intestine ranges from 0.15 to 0.3% (w/v) (Šušković et al. 2000). Any would-be probiotic must withstand these physiologically relevant concentrations of acid and bile in an in vitro assay for at least 3 h and 24 h, respectively (Bagci et al. 2019). NPL-493 withstood physiological acid concentrations for 3 h and bile for 24 h, consistent with other studies on *E. faecalis* (Zhang et al. 2016). High bile tolerance may be attributed to BSH activity, which reduces host cholesterol absorption (Bustos et al. 2018). BSH activity occurs widely in LAB species, including *Enterococcus* spp. and *E. faecalis,* with enhanced BSH activity (Chand et al. 2018) present therapeutic potential in mitigating hypercholesterolemia (Hernández-Gómez et al. 2021). The evolution of BSH function to survive intestinal bile (Foley et al. 2019) reinforces the enteromammary origins of the milk-derived *E. faecalis* strains isolated here. Oral probiotics also must survive digestion by enzymes present in the stomach and small intestine, and enterococci like NPL-493 can withstand trypsin and pepsin proteolysis (Lei et al. 2015). Antimicrobial capability is a highly desirable probiotic attribute empowering gut-resident bacteria to eliminate pathogenic GI infections (Franz et al. 2011) and modulate the gut microbiota to produce eubiosis (Carasi et al. 2017). Enterococci are crucial for protecting the neonatal gut from pathogens (Al Atya et al. 2015) such as *S. aureus, S. pyogenes*, *Salmonella* spp., and high-risk enterobacteria and ensuring the safety of donor human milk (Fernández et al. 2020). In agreement with our results, broad-spectrum anti-pathogen activity driven by lactic acid and bacteriocin-like inhibitory substances (BLIS) has been seen in *E. faecalis* strains from the newborn gut (Al Atya et al. 2015). The antimicrobial activity of NPL-493 and its harmlessness towards non-enterococcal LAB species confirms its position as the most suitable probiotic candidate among those evaluated here (Bagci et al. 2019).

Conclusions

Based on the results obtained in this study, we conclude that most *Enterococcus* strains have virulence potential and should be rejected as potential probiotics. However, NPL-493 exhibited beneficial in vitro and molecular traits in line with FAO/WHO criteria (FAO/WHO 2002). Combined with its relative avirulence, inability to cross the gut endothelial barrier, and absence of transferable antibiotic resistance, it comes very close to being the ideal enterococcal probiotic. However, further investigations into its ability to exchange DNA, its interaction with the human innate immune system, and infectivity in immunocompromised animals must be done before conclusively establishing its position as a premier human probiotic. Our study does reinforce the idea that an *Enterococcus* species carefully evaluated at the strain level can be a beneficial addition to supplements and foods for human consumption (Foulquié Moreno et al. 2006).

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Author contributions JA carried out all the experimental work and wrote the initial manuscript. AZ arranged for the ethical approvals, sampling of hospital patients, and logistics for fieldwork, edited and finalized the manuscript, supervised and provided guidance in experiment design and execution, and gave financial/administrative support of the overall research. KB supervised some of the lab experiments and resources. MT helped in the planning and validation of the experiments. All authors have approved the final manuscript.

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Availability of data and materials All data generated or analyzed during this study are included in this published article and its supplementary information files. Data about the DNA sequences mentioned in the manuscript are submitted in NCBI-GenBank.

Declarations

Conflict of interest None of the authors have any financial and non-financial competing interests to declare.

Ethical approval Research involving human subjects, human material, or human data presented here has been performed in accordance with the Declaration of Helsinki. It has been officially approved by the ethics committee of NIBGE and the Institutional review board (IRB) of the hospitals concerned.

Consent to participate Research involving human subjects, human material, or human data presented here has been performed in accordance with the Declaration of Helsinki.

Consent for publication Formal written consent from individuals reported here has been obtained to publish data in any form for research purposes.

References

Aarestrup FM, Agerso Y, Gerner-Smidt P, Madsen M, Jensen LB (2000) Comparison of antimicrobial resistance phenotypes and resistance genes in *Enterococcus faecalis* and *Enterococcus faecium* from humans in the community, broilers, and pigs in Denmark. Diagn Microbiol Infect Dis 37:127–137. https://doi.org/10.1016/S0732-8893(00)00130-9

Al Atya AK, Drider-Hadiouche K, Ravallec R, Silvain A, Vachee A, Drider D (2015) Probiotic potential of *Enterococcus faecalis* strains isolated from meconium. Front Microbiol 6:1–9. https://doi.org/10.3389/fmicb.2015.00227

Ali SA, Bin-Asif H, Hasan KA, Rehman M, Abbasi A (2017) Molecular assessment of virulence determinants, hospital associated marker (IS16gene) and prevalence of antibiotic resistance in soil borne *Enterococcus* species. Microb Pathog 105:298–306. https://doi.org/10.1016/j.micpath.2017.02.041

Andrighetto C et al (2001) Phenotypic and genetic diversity of enterococci isolated from Italian cheeses. J Dairy Res 68:303–316. https://doi.org/10.1017/S0022029901004800

Anjum J, Nazir S, Tariq M, Barrett K, Zaidi A (2020) *Lactobacillus* commensals autochthonous to human milk have the hallmarks of potent probiotics. Microbiol 166:966–980. https://doi.org/10.1099/mic.0.000966

Archambaud C, Derré-Bobillot A, Lapaque N, Rigottier-Gois L, Serror P (2019) Intestinal translocation of enterococci requires a threshold level of enterococcal overgrowth in the lumen. Sci Rep 9:1–12. https://doi.org/10.1038/s41598-019-45441-3

Arciola CR, Campoccia D, Baldassarri L, Pirini V, Huebner J, Montanaro L (2007) The role of *Enterococcus faecalis* in orthopaedic peri-implant infections demonstrated by automated ribotyping and cluster analysis. Biomaterials 28:3987–3995. https://doi.org/10.1016/j.biomaterials.2007.05.013

Arias CA, Murray BE (2012) The rise of the *Enterococcus*: beyond vancomycin resistance. Nat Rev Microbiol 10:266–278. https://doi.org/10.1038/nrmicro2761

Bagci U, Ozmen Togay S, Temiz A, Ay M (2019) Probiotic characteristics of bacteriocin-producing *Enterococcus faecium* strains isolated from human milk and colostrum. Folia Microbiol 64:735–750. https://doi.org/10.1007/s12223-019-00687-2

Barbieri F, Montanari C, Gardini F, Tabanelli G (2019) Biogenic amine production by lactic acid bacteria: a review. Foods 8:17

Belicová A, Mikulášová M, Dušinský R (2013) Probiotic potential and safety properties of *Lactobacillus plantarum* from Slovak Bryndza cheese. BioMed Res Int 2013:1–9. https://doi.org/10.1155/2013/760298

Blessing EN, Chukwuemeka IS, Ukeachu CD, Onuawuchi UG (2020) Antibacterial properties of probiotics bacterial isolated from human breast milk. World News Nat Sci 29:290–297

Bouridane H, Sifour M, Idoui T, Annick L, Thonard P (2016) Technological and probiotic traits of the lactobacilli isolated from vaginal tract of the healthy women for probiotic use. Iran J Biotechnol 14:192–201. https://doi.org/10.15171/ijb.1432

Buhnik-Rosenblau K, Matsko-Efimov V, Danin-Poleg Y, Franz CMAP, Klein G, Kashi Y (2013) Biodiversity of *Enterococcus faecalis* based on genomic typing. Int J Food Microbiol 165:27–34. https://doi.org/10.1016/j.ijfoodmicro.2013.04.009

Bustos AY, de Valdez GF, Fadda S, Taranto MP (2018) New insights into bacterial bile resistance mechanisms: the role of bile salt hydrolase and its impact on human health. Food Res Int 112:250–262

Carasi P, Racedo SM, Jacquot C, Elie AM, MdlÁ S, Urdaci MC (2017) *Enterococcus durans* EP1 a promising anti-inflammatory probiotic able to stimulate sIgA and to increase *Faecalibacterium prausnitzii* abundance. Front Immunol 8:1–14. https://doi.org/10.3389/fimmu.2017.00088

Casarotti SN, Carneiro BM, Todorov SD, Nero LA, Rahal P, Penna ALB (2017) In vitro assessment of safety and probiotic potential characteristics of *Lactobacillus* strains isolated from water buffalo mozzarella cheese. Ann Microbiol 67:289–301. https://doi.org/10.1007/s13213-017-1258-2

Cebrián R, Baños A, Valdivia E, Pérez-Pulido R, Martínez-Bueno M, Maqueda M (2012) Characterization of functional, safety, and probiotic properties of *Enterococcus faecalis* UGRA10, a new AS-48-producer strain. Food Microbiol 30:59–67. https://doi.org/10.1016/j.fm.2011.12.002

Chand D, Panigrahi P, Varshney N, Ramasamy S, Suresh CG (2018) Structure and function of a highly active bile salt hydrolase (BSH) from *Enterococcus faecalis* and post-translational processing of BSH enzymes. Biochim Biophys Acta, Proteins Proteom 1866:507–518. https://doi.org/10.1016/j.bbapap.2018.01.003

Christoffersen TE, Jensen H, Kleiveland CR, Dørum G, Jacobsen M, Lea T (2012) In vitro comparison of commensal, probiotic and pathogenic strains of *Enterococcus faecalis*. Br J Nutr 108:2043–2053. https://doi.org/10.1017/S0007114512000220

Clark CG (1997) Riboprinting: a tool for the study of genetic diversity in microorganisms. J Eukaryotic Microbiol 44:277–283. https://doi.org/10.1111/j.1550-7408.1997.tb05667.x

Comerlato CB, de Resende MCC, Caierão J, d'Azevedo PA (2013) Presence of virulence factors in *Enterococcus faecalis* and *Enterococcus faecium* susceptible and resistant to vancomycin. Mem Inst Oswaldo Cruz 108:590–595. https://doi.org/10.1590/S0074-02762013000500009

Creti R, Imperi M, Bertuccini L, Fabretti F, Orefici G, Di Rosa R, Baldassarri L (2004) Survey for virulence determinants among *Enterococcus faecalis* isolated from different sources. J Med Microbiol 53:13–20. https://doi.org/10.1099/jmm.0.05353-0

D'Costa VM et al (2011) Antibiotic resistance is ancient. Nature 477:457–461. https://doi.org/10.1038/nature10388

de Castilho NPA, Nero LA, Todorov SD (2019) Molecular screening of beneficial and safety determinants from bacteriocinogenic lactic acid bacteria isolated from Brazilian artisanal calabresa. Lett Appl Microbiol 69:204–211. https://doi.org/10.1111/lam.13194

de Souza BMS, Borgonovi TF, Casarotti SN, Todorov SD, Penna ALB (2019) *Lactobacillus casei* and *Lactobacillus fermentum* strains isolated from mozzarella cheese: probiotic potential, safety, acidifying kinetic parameters and viability under gastrointestinal tract conditions. Probiotics Antimicrob Proteins 11:382–396. https://doi.org/10.1007/s12602-018-9406-y

De las Rivas B, Marcobal Á, Muñoz R (2005) Improved multiplex-PCR method for the simultaneous detection of food bacteria producing biogenic amines. FEMS Microbiol Lett 244:367–372. https://doi.org/10.1016/j.femsle.2005.02.012

De Leener E, Martel A, Decostere A, Haesebrouck F (2004) Distribution of the erm (B) gene, tetracycline resistance genes, and Tn1545-like transposons in macrolide- and lincosamide-resistant enterococci from pigs and humans. Microb Drug Resist 10:341–345. https://doi.org/10.1089/mdr.2004.10.341

Devriese S et al (2017) T84 monolayers are superior to Caco-2 as a model system of colonocytes. Histochem Cell Biol 148:85–93. https://doi.org/10.1007/s00418-017-1539-7

Dubin K, Pamer EG (2018) Enterococci and their interactions with the intestinal microbiome. In: Britton RA, Cani PD (eds) Bugs as drugs: therapeutic microbes for the prevention and treatment of disease. ASM press, Washington DC, USA, pp 309–330

Dundar H, Brede DA, La Rosa SL, El-Gendy AO, Diep DB, Nes IF (2015) The fsr quorum-sensing system and cognate gelatinase orchestrate the expression and processing of proprotein ef_1097 into the mature antimicrobial peptide enterocin o16. J Bacteriol 197:2112–2121. https://doi.org/10.1128/JB.02513-14

Dutka-Malen S, Evers S, Courvalin P (1995) Detection of glycopeptide resistance genotypes and identification to the species level of clinically relevant enterococci by PCR. J Clin Microbiol 33:24–27. https://doi.org/10.1128/jcm.33.1.24-27.1995

Eaton TJ, Gasson MJ (2001) Molecular screening of *Enterococcus* virulence determinants and potential for genetic exchange between food and medical isolates. Appl Env Microbiol 67:1628–1635. https://doi.org/10.1128/AEM.67.4.1628-1635.2001

EFSA-Panel (2011) Scientific opinion on risk based control of biogenic amine formation in fermented foods. EFSA J 9(10):1–93. https://doi.org/10.2903/j.efsa.2011.2393

EFSA-Panel (2012) Guidance on the safety assessment of *Enterococcus faecium* in animal nutrition. EFSA J 10:1–10. https://doi.org/10.2903/j.efsa.2012.2682

Ejaz H (2019) Emerging resistance of van genotype in enterococci: a potential menace for therapeutic failure. Pak J Med Sci 35:1659–1663. https://doi.org/10.12669/pjms.35.6.1145

FAO/WHO (2002) Report of a joint FAO/WHO working group on drafting guidelines for the evaluation of probiotics in food. FAO/WHO, London Ontario, Canada

Faron ML, Ledeboer NA, Buchan BW (2016) Resistance mechanisms, epidemiology, and approaches to screening for vancomycin-resistant *Enterococcus* in the health care setting. J Clin Microbiol 54:2436. https://doi.org/10.1128/JCM.00211-16

Fernández L, Langa S, Martín V, Maldonado A, Jiménez E, Martín R, Rodríguez JM (2013) The human milk microbiota: origin and potential roles in health and disease. Pharmacol Res 69:1–10. https://doi.org/10.1016/j.phrs.2012.09.001

Fernández L, Pannaraj PS, Rautava S, Rodríguez JM (2020) The microbiota of the human mammary ecosystem. Front Cell Infect Microbiol 10:586667–586667. https://doi.org/10.3389/fcimb.2020.586667

Fisher K, Phillips C (2009) The ecology, epidemiology and virulence of *Enterococcus*. Microbiol 155:1749–1757. https://doi.org/10.1099/mic.0.026385-0

Foley MH, O'Flaherty S, Barrangou R, Theriot CM (2019) Bile salt hydrolases: gatekeepers of bile acid metabolism and host-microbiome crosstalk in the gastrointestinal tract. PLoS Pathog 15:e1007581. https://doi.org/10.1371/journal.ppat.1007581

Fortina M et al (2008) A survey on biotechnological potential and safety of the novel *Enterococcus* species of dairy origin, *E Italicus*. Int J Food Microbiol 123:204–211. https://doi.org/10.1016/j.ijfoodmicro.2008.01.014

Foulquié Moreno MR, Sarantinopoulos P, Tsakalidou E, De Vuyst L (2006) The role and application of enterococci in food and health. Int J Food Microbiol 106:1–24. https://doi.org/10.1016/j.ijfoodmicro.2005.06.026

Franz CM, Huch M, Abriouel H, Holzapfel W, Gálvez A (2011) Enterococci as probiotics and their implications in food safety. Int J Food Microbiol 151:125–140. https://doi.org/10.1016/j.ijfoodmicro.2011.08.014

Galloway-Pena JR, Bourgogne A, Qin X, Murray BE (2011) Diversity of the fsr-gelE region of the *Enterococcus faecalis* genome but conservation in strains with partial deletions of the fsr operon. Appl Environ Microbiol 77:442–451. https://doi.org/10.1128/AEM.00756-10

Gardini F, Bover-Cid S, Tofalo R, Belletti N, Gatto V, Suzzi G, Torriani S (2008) Modeling the aminogenic potential of *Enterococcus faecalis* EF37 in dry fermented sausages through chemical and molecular approaches. Appl Environ Microbiol 74:2740–2750. https://doi.org/10.1128/aem.02267-07

Georgieva R et al (2015) Antimicrobial activity and antibiotic susceptibility of *Lactobacillus* and *Bifidobacterium* spp. intended for use as starter and probiotic cultures. Biotechnol Biotechnol Equip 29:84–91. https://doi.org/10.1080/13102818.2014.987450

Gevers D, Danielsen M, Huys G, Swings J (2003) Molecular characterization of tet (M) genes in *Lactobacillus* isolates from different types of fermented dry sausage. Appl Env Microbiol 69:1270–1275. https://doi.org/10.1128/AEM.69.2.1270-1275.2003

Golob M, Pate M, Kušar D, Dermota U, Avberšek J, Papić B, Zdovc I (2019) Antimicrobial resistance and virulence genes in *Enterococcus faecium* and *Enterococcus faecalis* from humans and retail red meat. BioMed Res Int 2019:1–12. https://doi.org/10.1155/2019/2815279

Gómez NC, Ramiro JM, Quecan BX, de Melo Franco BD (2016) Use of potential probiotic lactic acid bacteria (LAB) biofilms for the control of *Listeria monocytogenes, Salmonella Typhimurium*, and *Escherichia coli* O157: H7 biofilms formation. Front Microbiol 7:863. https://doi.org/10.3389/fmicb.2016.00863

Gueimonde M, Sánchez B, de Los Reyes-Gavilán CG, Margolles A (2013) Antibiotic resistance in probiotic bacteria. Front Microbiol 4:202. https://doi.org/10.3389/fmicb.2013.00202

Hanchi H, Mottawea W, Sebei K, Hammami R (2018) The genus *Enterococcus*: between probiotic potential and safety concerns—an update. Front Microbiol 9:1–16. https://doi.org/10.3389/fmicb.2018.01791

Hasan KA, Ali SA, Rehman M, Bin-Asif H, Zahid S (2018) The unravelled *Enterococcus faecalis* zoonotic superbugs: emerging multiple resistant and virulent lineages isolated from poultry environment. Zoonoses Public Health 65:921–935. https://doi.org/10.1111/zph.12512

Hernández-Gómez JG, López-Bonilla A, Trejo-Tapia G, Ávila-Reyes SV, Jiménez-Aparicio AR, Hernández-Sánchez H (2021) In vitro bile salt hydrolase (BSH) activity screening of different probiotic microorganisms. Foods 10:1–10. https://doi.org/10.3390/foods10030674

Higuita NIA, Huycke MM (2014) Enterococcal disease, epidemiology, and implications for treatment. In: Gilmore MSCD, Ike Y, Nathan S (eds) Enterococci: from commensals to leading causes of drug resistant infection [Internet]. Massachusetts Eye and Ear Infirmary, Boston, Massachuesetts USA

Hollenbeck BL, Rice LB (2012) Intrinsic and acquired resistance mechanisms in *Enterococcus*. Virulence 3:421–569. https://doi.org/10.4161/viru.21282

Huang MS, Cheng CC, Tseng SY, Lin YL, Hm Lo, Chen PW (2019) Most commensally bacterial strains in human milk of healthy mothers display multiple antibiotic resistance. Microbiol Open 8:e00618. https://doi.org/10.1002/mbo3.618

Hummel A, Holzapfel WH, Franz CMAP (2007) Characterisation and transfer of antibiotic resistance genes from enterococci isolated from food. Sys Appl Microbiol 30:1–7. https://doi.org/10.1016/j.syapm.2006.02.004

Inoğlu ZN, Tuncer Y (2013) Safety assessment of *Enterococcus faecium* and *Enterococcus faecalis* strains isolated from Turkish Tulum cheese. J Food Saf 33:369–377. https://doi.org/10.1111/jfs.12061

Jahan M, Zhanel GG, Sparling R, Holley RA (2015) Horizontal transfer of antibiotic resistance from *Enterococcus faecium* of fermented meat origin to clinical isolates of *E. faecium* and *Enterococcus faecalis*. Int J Food Microbiol 199:78–85. https://doi.org/10.1016/j.ijfoodmicro.2015.01.013

James KR et al (2020) Distinct microbial and immune niches of the human colon. Nat Immunol 21:343–353. https://doi.org/10.1038/s41590-020-0602-z

Jiménez E, Delgado S, Fernández L, García N, Albújar M, Gómez A, Rodríguez JM (2008) Assessment of the bacterial diversity of human colostrum and screening of staphylococcal and enterococcal populations for potential virulence factors. Res Microbiol 159:595–601. https://doi.org/10.1016/j.resmic.2008.09.001

Jiménez E et al (2013) Antibiotic resistance, virulence determinants and production of biogenic amines among enterococci from ovine, feline, canine, porcine and human milk. BMC Microbiol 13:1–12. https://doi.org/10.1186/1471-2180-13-288

Kayser F (2003) Safety aspects of enterococci from the medical point of view. Int J Food Microbiol 88:255–262. https://doi.org/10.1016/S0168-1605(03)00188-0

Khalkhali S, Mojgani N (2017) Bacteriocinogenic potential and virulence traits of *Enterococcus faecium* and *E. faecalis* isolated from human milk. Iran J Microbiol 9:224–233

Khan Z, Siddiqui N, Saif MW (2018) *Enterococcus faecalis* infective endocarditis and colorectal carcinoma: case of new association gaining ground. Gastroenterol Res 11:238–240. https://doi.org/10.14740/gr996w

Kıvanç SA, Kıvanç M, Yiğit T (2016) Antibiotic susceptibility, antibacterial activity and characterisation of *Enterococcus faecium* strains isolated from breast milk. Exp Ther Med 12:1732–1740. https://doi.org/10.3892/etm.2016.3545

Korpela K et al (2018) Fucosylated oligosaccharides in mother's milk alleviate the effects of caesarean birth on infant gut microbiota. Sci Rep 8:1–7. https://doi.org/10.1038/s41598-018-32037-6

Kristich CJRL, Arias CA (2014) Enterococcal infection—treatment and antibiotic resistance. In: Gilmore MSCD, Ike Y, Shankar N (eds) Enterococci: from commensals to leading causes of drug resistant infection. Massachusetts Eye and Ear Infirmary, Boston, Massachussets, USA, pp 123–184

Kvist LJ, Larsson BW, Hall-Lord ML, Steen A, Schalén C (2008) The role of bacteria in lactational mastitis and some considerations of the use of antibiotic treatment. Int Breastfeed J 3:1–7. https://doi.org/10.1186/1746-4358-3-6

Landete JM, Peirotén Á, Medina M, Arqués JL, Rodríguez-Mínguez E (2018) Virulence and antibiotic resistance of enterococci isolated from healthy breastfed infants. Microb Drug Resist 24:63–69. https://doi.org/10.1089/mdr.2016.0320

Laursen MF et al (2020) Intestinal *Enterococcus* abundance correlates inversely with excessive weight gain and increased plasma leptin in breastfed infants. FEMS Microbiol Ecol 96:1–9. https://doi.org/10.1093/femsec/fiaa066

Lei M, Dai X, Liu M (2015) Biological characteristics and safety examination of five enterococcal strains from probiotic products. J Food Saf 35:324–335. https://doi.org/10.1111/jfs.12179

Li W, Raoult D, Fournier P-E (2009) Bacterial strain typing in the genomic era. FEMS Microbiol Rev 33:892–916. https://doi.org/10.1111/j.1574-6976.2009.00182.x

Li B et al (2018) Evaluating the safety of potential probiotic *Enterococcus durans* KLDS6.0930 using whole genome sequencing and oral toxicity study. Front Microbiol 9:1–15. https://doi.org/10.3389/fmicb.2018.01943

Li Y, Shan L, Zhang C, Lei Z, Shang Y (2019) Isolation and antibiotic resistant research of *Tetragenococcus halophilus* from Xuanwei Ham, a China high-salt-fermented meat products. Antibiotics 8:1–16. https://doi.org/10.3390/antibiotics8030151

Liong M-T (2008) Safety of probiotics: translocation and infection. Nutr Rev 66:192–202. https://doi.org/10.1111/j.1753-4887.2008.00024.x

Lopes MDFS, Simões AP, Tenreiro R, Marques JJF, Crespo MTB (2006) Activity and expression of a virulence factor, gelatinase, in dairy enterococci. Int J Food Microbiol 112:208–214. https://doi.org/10.1016/j.ijfoodmicro.2006.09.004

Manson JM, Keis S, Smith JM, Cook GM (2004) Acquired bacitracin resistance in *Enterococcus faecalis* is mediated by an ABC transporter and a novel regulatory protein, BcrR. Antimicrob Agents Chemother 48:3743–3748. https://doi.org/10.1128/AAC.48.10.3743-3748.2004

Martín-Platero AM, Valdivia E, Maqueda M, Martínez-Bueno M (2009) Characterization and safety evaluation of enterococci isolated from Spanish goats' milk cheeses. Int J Food Microbiol 132:24–32. https://doi.org/10.1016/j.ijfoodmicro.2009.03.010

Miele A, Bandera M, Goldstein BP (1995) Use of primers selective for vancomycin resistance genes to determine van genotype in enterococci and to study gene organization in VanA isolates. Antimicrob Agents Chemother 39:1772–1778. https://doi.org/10.1128/AAC.39.8.1772

Monteagudo-Mera A, Caro I, Rodriguez-Aparicio L, Rúa J, Ferrero M, García-Armesto M (2011) Characterization of certain bacterial strains for potential use as starter or probiotic cultures in dairy products. J Food Prot 74:1379–1386. https://doi.org/10.4315/0362-028X.JFP-10-392

Naila A, Flint S, Fletcher G, Bremer P, Meerdink G (2010) Control of biogenic amines in food-existing and emerging approaches. J Food Sci 75:R139-150. https://doi.org/10.1111/j.1750-3841.2010.01774.x

Nueno-Palop C, Narbad A (2011) Probiotic assessment of *Enterococcus faecalis* CP58 isolated from human gut. Int J Food Microbiol 145:390–394. https://doi.org/10.1016/j.ijfoodmicro.2010.12.029

Perez M, Calles-Enríquez M, Nes I, Martin MC, Fernandez M, Ladero V, Alvarez MA (2015) Tyramine biosynthesis is transcriptionally induced at low pH and improves the fitness of *Enterococcus faecalis* in acidic environments. Appl Microbiol Biotechnol 99:3547–3558. https://doi.org/10.1007/s00253-014-6301-7

Perez M et al (2016) Mastitis modifies the biogenic amines profile in human milk, with significant changes in the presence of histamine, putrescine and spermine. PLoS ONE 11:1–10. https://doi.org/10.1371/journal.pone.0162426

Peta M et al (2006) Outbreak of vancomycin-resistant *Enterococcus* spp. in an Italian general intensive care unit. Clin Microbiol Inf 12:163–169. https://doi.org/10.1111/j.1469-0691.2005.01331.x

Portela CA, Smart KF, Tumanov S, Cook GM, Villas-Bôas SG (2014) Global metabolic response of *Enterococcus faecalis* to oxygen. J Bacteriol 196:2012–2022. https://doi.org/10.1128/JB.01354-13

Ramiah K, Van Reenen C, Dicks L (2007) Expression of the mucus adhesion genes Mub and MapA, adhesion-like factor EF-Tu and bacteriocin gene plaA of *Lactobacillus plantarum* 423, monitored with real-time PCR. Int J Food Microbiol 116:405–409. https://doi.org/10.1016/j.ijfoodmicro.2007.02.011

Reis N, Saraiva M, Duarte E, de Carvalho E, Vieira B, Evangelista-Barreto N (2016) Probiotic properties of lactic acid bacteria isolated from human milk. J Appl Microbiol 121:811–820. https://doi.org/10.1111/jam.13173

Reviriego C, Eaton T, Martín R, Jiménez E, Fernández L, Gasson MJ, Rodríguez JM (2005) Screening of virulence determinants in *Enterococcus faecium* strains isolated from breast milk. J Hum Lact 21:131–137. https://doi.org/10.1177/0890334405275394

Rodríguez JM (2014) The origin of human milk bacteria: is there a bacterial entero-mammary pathway during late pregnancy and lactation? Adv Nutr 5:779–784. https://doi.org/10.3945/an.114.007229

Ruiz-Garbajosa P et al (2009) Longer intestinal persistence of *Enterococcus faecalis* compared to *Enterococcus faecium* clones in intensive-care-unit patients. J Clin Microbiol 47:345. https://doi.org/10.1128/JCM.01597-08

Santana LAM et al (2020) Identification and characterization of resistance and pathogenicity of *Enterococcus* spp. in samples of donor breast milk. World J Clin Pediatr 9:53–62. https://doi.org/10.5409/wjcp.v9.i3.53

Sartingen S, Rozdzinski E, Muscholl-Silberhorn A, Marre R (2000) Aggregation substance increases adherence and internalization, but not translocation, of *Enterococcus faecalis* through different intestinal epithelial cells in vitro. Inf Immun 68:6044–6047. https://doi.org/10.1128/IAI.68.10.6044-6047.2000

Sedláčková P, Horáčková Š, Shi T, Kosova M, Plocková M (2015) Two different methods for screening of bile salt hydrolase activity in *Lactobacillus* strains. Czech J Food Sci 33:13–18. https://doi.org/10.17221/299/2014-CJFS

Shi Y, Zhai M, Li J, Li B (2020) Evaluation of safety and probiotic properties of a strain of *Enterococcus faecium* isolated from chicken bile. J Food Sci Technol 57:578–587

Sirichoat A, Flórez AB, Vázquez L, Buppasiri P, Panya M, Lulitanond V, Mayo B (2020) Antibiotic resistance-susceptibility profiles of *Enterococcus faecalis* and *Streptococcus* spp. from the human vagina, and genome analysis of the genetic basis of intrinsic and acquired resistances. Front Microbiol 11:1–10. https://doi.org/10.3389/fmicb.2020.01438

Steck N et al (2011) *Enterococcus faecalis* metalloprotease compromises epithelial barrier and contributes to intestinal inflammation. Gastroenterology 141:959–971. https://doi.org/10.1053/j.gastro.2011.05.035

Šušković J, Kos B, Matošić S, Besendorfer V (2000) The effect of bile salts on survival and morphology of a potential probiotic strain *Lactobacillus acidophilus* M92. World J Microbiol Biotechnol 16:673–678. https://doi.org/10.1023/A:1008909505651

Sutcliffe J, Grebe T, Tait-Kamradt A, Wondrack L (1996) Detection of erythromycin-resistant determinants by PCR. Antimicrob Agents Chemother 40:2562–2566. https://doi.org/10.1128/AAC.40.11.2562

Taweerodjanakarn S, Haertlé T, Chobert J-M, Hongpattarakere T (2019) Functional properties of *Enterococcus faecalis* isolated from colostrum drawn from Thai mothers. Int Food Res J 26:141–151

Teixeira N et al (2012) The incongruent gelatinase genotype and phenotype in *Enterococcus faecalis* are due to shutting off the ability to respond to the gelatinase biosynthesis-activating pheromone (GBAP) quorum-sensing signal. Microbiol 158:519–528. https://doi.org/10.1099/mic.0.055574-0

Toğay SÖ, Temiz A, Celebi A, Acik L, Yalçin SS (2014) Investigation of potential virulence genes and antibiotic resistance characteristics of *Enterococcus faecalis* isolates from human milk and colostrum samples. Turk J Biol 38:357–364. https://doi.org/10.3906/biy-1311-34

Tsikrikonis G et al (2012) Differences in biofilm formation and virulence factors between clinical and fecal enterococcal isolates of human and animal origin. Microb Pathog 52:336–343. https://doi.org/10.1016/j.micpath.2012.03.003

Valenzuela AS, Omar NB, Abriouel H, López RL, Ortega E, Cañamero MM, Gálvez A (2008) Risk factors in enterococci isolated from foods in Morocco: determination of antimicrobial resistance and incidence of virulence traits. Food Chem Toxicol 46:2648–2652. https://doi.org/10.1016/j.fct.2008.04.021

Valenzuela AS et al (2009) Virulence factors, antibiotic resistance, and bacteriocins in enterococci from artisan foods of animal origin. Food Control 20:381–385. https://doi.org/10.1016/j.foodcont.2008.06.004

Vankerckhoven V et al (2004) Development of a multiplex PCR for the detection of *asa1, gelE, cylA, esp*, and *hyl* genes in enterococci and survey for virulence determinants among European hospital isolates of *Enterococcus faecium*. J Clin Microbiol 42:4473–4479. https://doi.org/10.1128/JCM.42.10.4473-4479.2004

Wells CL, Jechorek RP, Erlandsen SL (1990) Evidence for the translocation of *Enterococcus faecalis* across the mouse intestinal tract. J Infect Dis 162:82–90. https://doi.org/10.1093/infdis/162.1.82

Yamaguchi T, Miura Y, Matsumoto T (2013) Antimicrobial susceptibility of *Enterococcus* strains used in clinical practice as probiotics. J Infect Chemother 19:1109–1115. https://doi.org/10.1007/s10156-013-0633-6

Zeng J, Teng F, Weinstock GM, Murray BE (2004) Translocation of *Enterococcus faecalis* strains across a monolayer of polarized human enterocyte-like T84 cells. J Clin Microbiol 42:1149–1154. https://doi.org/10.1128/JCM.42.3.1149-1154.2004

Zhang F et al (2016) Screening probiotic strains for safety: evaluation of virulence and antimicrobial susceptibility of enterococci from healthy Chinese infants. J Dairy Sci 99:4282–4290. https://doi.org/10.3168/jds.2015-10690

Zielińska D, Kolożyn-Krajewska D (2018) Food-origin lactic acid bacteria may exhibit probiotic properties. BioMed Res Int 2018:1–15. https://doi.org/10.1155/2018/5063185